

What is an Industrial Manipulator

An Industrial Manipulator is an Ergonomic Lift Assist Device to help workers lift, manoeuvre and position products or loads that are too heavy, too awkward, or too dangerous to be done manually. It enables the operator to do all the manual tasks they did before but without physical effort and without physical risks.

Industrial Manipulators have Rigid Articulating Arms with terminal arms and end-effectors suitable for handling standard or offset loads and to move all loads in all directions in a weightless balanced condition and with movement in direct proportion to that of the operator. This makes them easier to use, quicker to use, less tiring for the operators over the total shift with no risk of injury to operatives or damage to the product.

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€ Industrial manipulators differ from cranes and hoists in that cranes and hoists can only move loads up and down in spaces where there are no overhead obstructions. By contrast, industrial manipulators with their rigid articulating arms do not have these limitations and can not only manoeuvre in any direction but can reach under obstructions or into enclosed spaces.

€ Industrial Manipulators differ from robots in that robots are fully programmed to operate repetitively without any physical human interaction. They are beneficial for large production runs where all parts and movements in the process are predictable and programmable and which can cost justify the higher purchase costs and higher ongoing maintenance costs. By contrast, the industrial manipulator is a lift assist device for the human operator which works together with the operator. But removes from the operator the physical effort of the task and removes any risk of injury to the operator or to the product.

The uses of Industrial Manipulators in the industry are virtually unlimited

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- They can Lift, Turn, Tip, Place, Incline, Position, Handle, Grip, Stack, Rotate, Tilt, Empty, Transfer, and more.
- The range of available features enables full customisation for almost any lifting task of any product.
- They are used or can be configured to be suitable for use in almost any manufacturing, assembly, or distribution operation.
- Just some of the many end-effector / gripping methods include Open Hook, Closed Hook, Pail Hook, Fork, Cradle, Mechanical or Pneumatic Outer Gripping Jaws, Inner Gripping Jaws, Mechanical Pentagraphs, Pneumatic Inner Mandrels, Suction Cups using Venturi Suction or Vacuum Generation and many, many more. They can be fixed, adjustable, or interchangeable as required.

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The Armtec Series of Industrial Manipulators include different configurations but all using the same industrial manipulator technology:

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- They use only compressed air of between 7bar to 9bar to grip, balance, lift, move, manoeuvre and position any product or load weighing between 10kg and 1,000kg in a weightless, zero gravity condition.
- They have air reservoirs, one-way valves, and other safety and operating features that prevent the manipulator from releasing the load until it is on a stable surface, even if there is a loss of air to the system or even in the event the operator accidentally gives the wrong command.
- They have a system of two distinct pneumatic circuits. The first circuit balances the weight of the arm and tooling (‘no-load’ balancing). The second circuit balances the arm, the tooling, and the load to be lifted (‘load’ balance).
- They come with a choice of weight balancing systems such as Weight Pre-selector, Up/Down Lever, Regulator, Auto balancing which can be further customised with other more specialised balancing options.
- They can be supplied as a fully integrated unit specifically designed to suit your application (product, operation, and work area). The manipulator size, balancing cylinder, length and shape of arm, and tooling design will all be ergonomically integrated for that task. Customers can also choose to buy the manipulator as a flange unit onto which the customer can add its own end-effector tooling.
- The manipulator can be configured to be column-mounted, overhead fixed, overhead trolley mounted to run on the overhead rail; or mounted on a forkable base plate; or pallet truck.
- The manipulator body has a 360-degree rotation at the mounting axis and a 110-degree rotation at the terminal arm and comes with Up/Down and Rotation Parking Brakes and many other operating and safety features.

The Armtec Series of Industrial Manipulators are Pneumatic Industrial Manipulators and use a system of two distinct pneumatic circuits. This makes them easier to use, safer to use, and allows for more operating features and safety features.

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Armtec can also supply the BA100 Cable Arm manipulator. This manipulator is limited to the up-down movement to a maximum 100kg lifting weight but comes with a jointed double arm for easy movement and has the advantages of the ease of use and balancing effect of the industrial manipulator and with all the safety features of the industrial manipulator.

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Other benefits of choosing Armtec Pneumatic Manipulators

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- Armtec manipulators are CE certified. They are designed and manufactured to BS EN ISO 12100:2010 Standards and to comply with the European 2006/42/EC Machinery Safety Directive (MD). This means the manipulator is designed based on existing directives and laws on the safety of machinery, pressure equipment, principles for risk assessment.
- Armtec test each industrial manipulator in simulated conditions before dismantling and delivery. If samples are supplied they can do full testing and guarantee performance on your product without the need for any on-site adjustments.
- Armtec Manipulators come with a 12 months warranty and a detailed operation and maintenance manual.
- Armtec uses pneumatic components supplied by quality companies who have representation worldwide. (SMC,
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Norgren, Festo, Parker, and similar). These parts are detailed in the Operating Manual and can be purchased directly from the supplier or through Posilift. Armtec also uses their own proprietary parts which are available through Posilift. This is a truly amazing technology that only uses 6bar to 9 bar compressed air as a fuel source.

Posilift skill is its ability to customise pneumatic industrial manipulator technology to best solve the materials handling problem of its customers and to specify the equipment that best meets their needs and budget.

Posilift has been specifying and selling pneumatic industrial manipulators since 1990. During its 30 years of operation, Posilift has adapted and customised pneumatic industrial manipulator technology to meet the changing needs of industry and the changing Health and Safety regulations around the world.

Industrial Manipulators still remain the best and most cost-effective lift assist device solution for the repetitive handling of heavy, awkward, offset, or dangerous loads.

Posilift is a worldwide distributor of Armttec Pneumatic Industrial Manipulators.

[Contact Us](#) to Get a Customised solution. Call: **+61 2 9415 1914**